Smart Rings: The Future of Wearables?

Smart rings are one of the fastest-growing segments of the wearable technology market. By 2026, smart rings will no longer be a novelty, but a fully-fledged alternative to smartwatches and fitness trackers. These compact gadgets successfully combine the style of jewelry with health monitoring features. They are popular among those who want to track sleep, activity, and recovery without the need for bulky wrist devices. The smart ring market is growing at an impressive rate: it is predicted to reach hundreds of millions of dollars by the 2030s, thanks to brands like **Oura**, **Samsung**, **Ultrahuman**, as well as numerous budget-friendly *Chinese clones*.



I won't dwell on the design of these devices, but I will simply note that unlike smartwatches, which often look like gadgets, rings easily fit into everyday style and can be worn as ordinary jewelry, without significantly burdening the wearer. Top-end models are made of *titanium*, *ceramic*, or *steel* and are not cheap. Budget brands offer a similar look for much less, making the technology accessible to a mass audience.

Advantages of smart rings

- Relative comfort: the ring doesn't interfere with sleep or exercise, maintaining constant contact with the skin for accurate measurements.
- Health monitoring: sleep tracking (stages and quality), heart rate, heart rate variability (HRV), blood oxygen level (SpO_2), body temperature, steps, and stress.
- Decent battery life: the device lasts 5-7 days.

Disadvantages

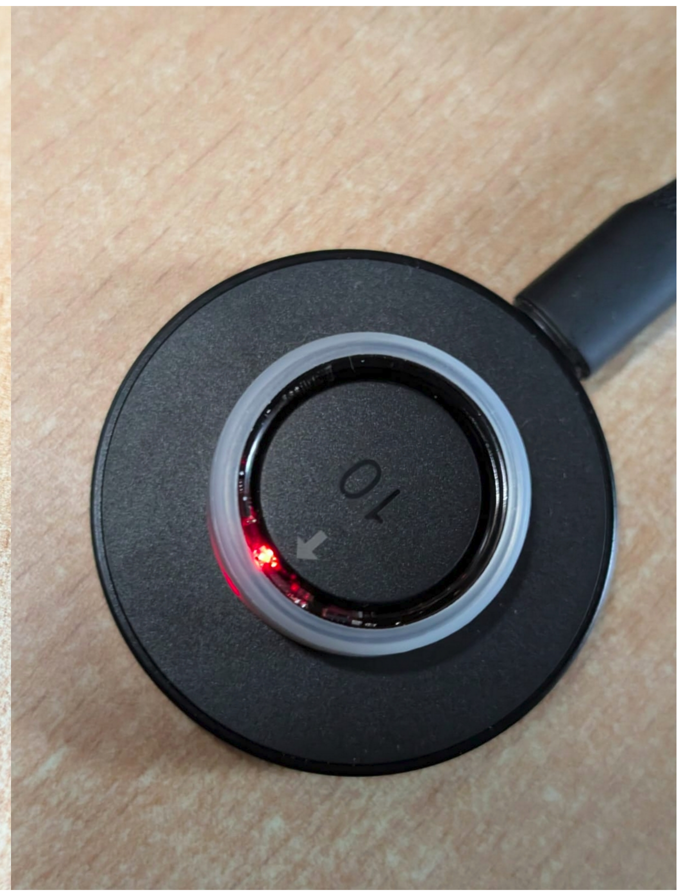
- Limited functionality: no notification screen (except for rare models).
- Accuracy: budget models may have inaccuracies in data, especially compared to premium devices.
- Price: Premium models like **Oura** or **Samsung** can cost up to $300-$500.
- Smartphone dependence: data is viewed in an app.

What can a budget smart ring do?

To understand what a smart ring can do, you need to at least understand what data it can provide and how to access it. Also, if you're not a millionaire, it's important to determine which smart ring model to choose to satisfy your curiosity without breaking the bank. Open source *Gadgetbridge*, which specializes in reverse engineering, easily suggested the choice: budget smart rings **COLMI**. Just $25 and a ton of experiences for the next couple of weeks or months.

I chose two rings for experimentation: the **R09** from 2024 and the **R12** from 2025. They differ slightly in appearance and functionality. Judging by the advertising and descriptions, both rings measure **heart rate**, **saturation (SpO_2)**, and **HRV**, detect the **stress**, and also analyze *sleep* and *physical activity*. The **R12** has a semblance of a display on the outside of the ring, but the **R09** can measure **temperature**.

COLMI R09



COLMI R12

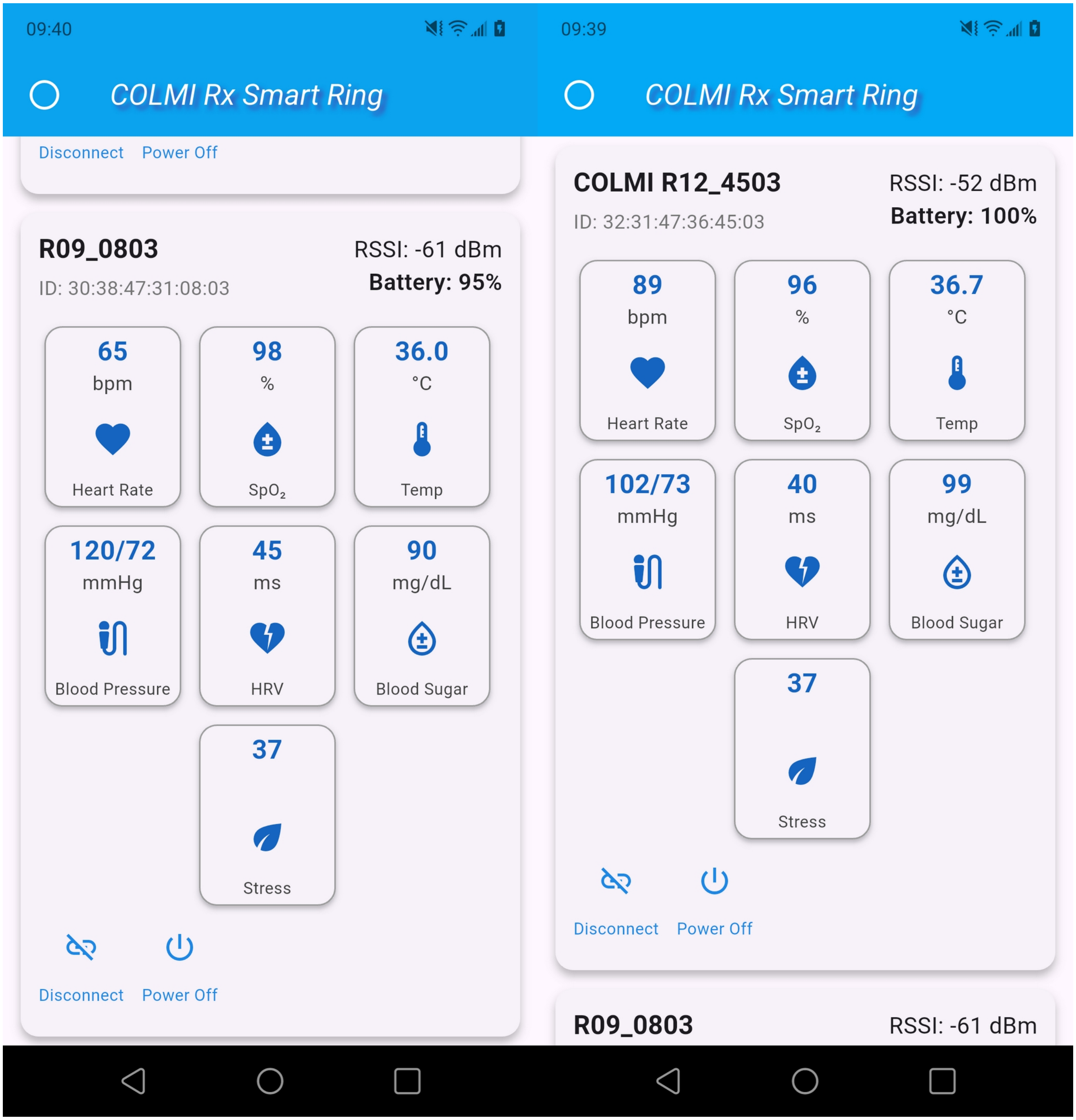
Research

I must say that the open source *Gadgetbridge* was a major disappointment. Although the ***QRing*** app that came with the **COLMI** smart rings clearly allowed for standard manual measurements, *Gadgetbridge* limited itself to describing only the heart rate measurement command. However, a huge amount of effort was invested in the code extracting information from the ring about physical activity, sleep duration, and continuous *heart rate* monitoring data, *SpO_2*, *HRV*, and *stress*. In other words into theme, something that doesn't interest me at all.

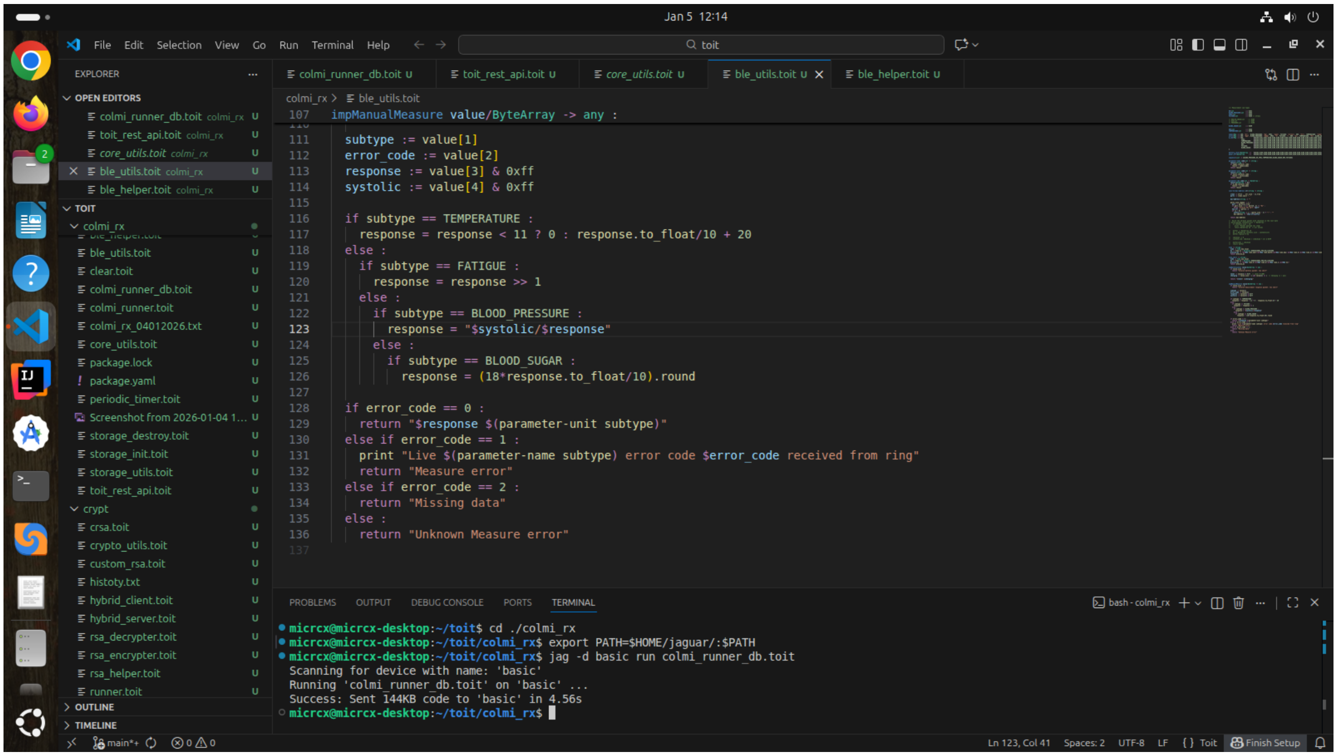
I had to search further and found a two-year-old project written in **Python**, *COLMI R02 CLIENT*. Functionally, this project is no better than *Gadgetbridge*, and I had a sneaking suspicion that *Gadgetbridge* had simply ported the *Python* code to *Java* and incorporated it into their project. However, in one of the project files (*real_time.py*), I found a hint on how to measure *SpO_2*, *HRV*, and *stress*. It turned out that the heart rate measurement code described in *Gadgetbridge* is just a special case of the general manual measurement system, and three additional parameters can also be measured. Moreover, assuming that the command table could be expanded, I was able to reconstruct the *temperature* measurement method and also discovered hidden, never-advertised *heart pressure* measurement commands and something completely exotic: *blood sugar*. In other words, the ***R09*** ring suddenly transformed from a *training aircraft* into a fifth-generation *F-35I monster fighter*. For $25, you can easily measure seven vital body parameters.

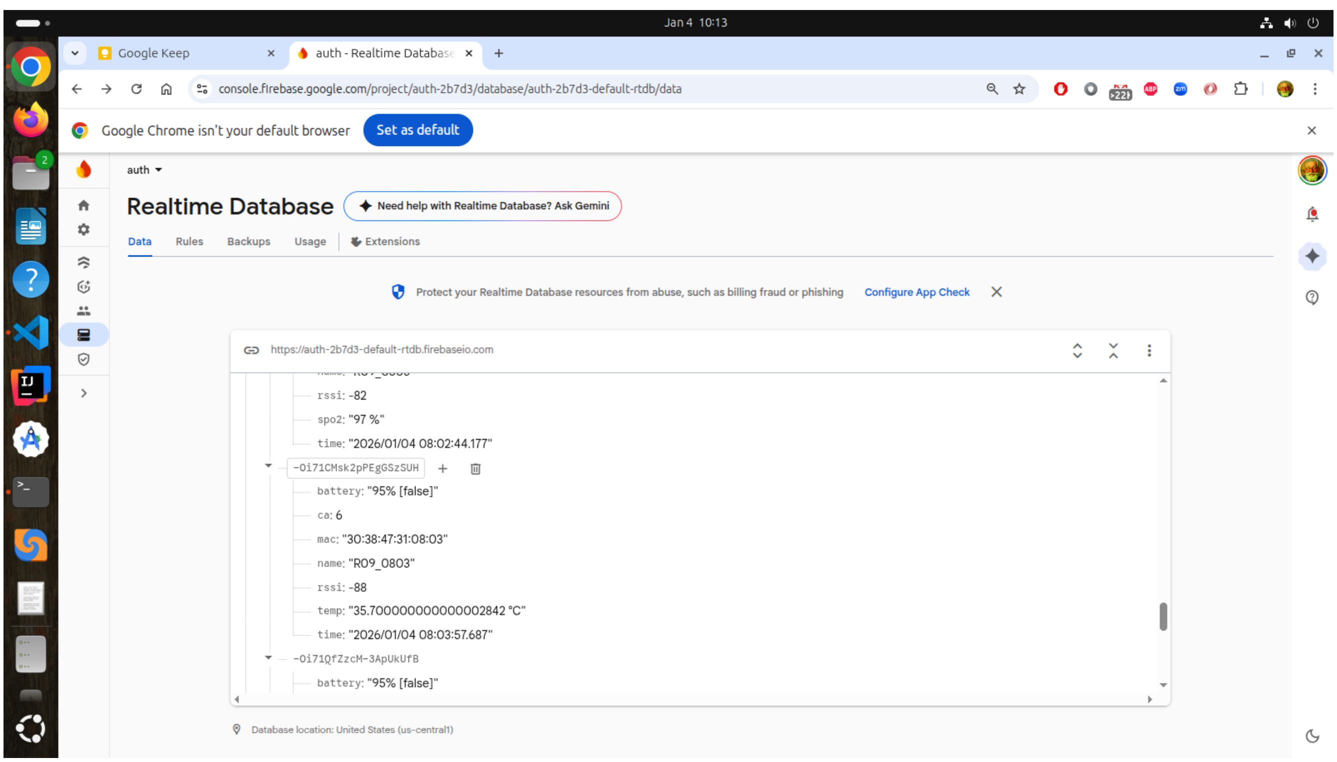
The next discovery was the fact that rings ***R09*** and ***R12*** are identical in their functionality, and the marketing descriptions of these electronic products are pure nonsense. I dare to suggest that rings ***R10*** and ***R11*** could be added to this group.

I wrote a small test app for *Android*, using *Dart* and *Flutter*, that measures some body parameters using rings ***R09*** and ***R12***. The app is a simple software gateway and allows simultaneous access to multiple rings for measurement, as shown in the image below:



Furthermore, I wrote an app for the **ESP32** that allows for a series of measurements of the aforementioned seven parameters and saves the data to the cloud. This resolves the question ([*LinkedIn*]) of whether it's possible to implement a gateway on the **ESP32** using **Toit**, a *high-level managed language*. It's possible, no doubt about it.


Jan 5 12:14
File Edit Selection View Go Run Terminal Help
EXPLORER
OPEN EDITORS
colmi_runner_db.toit
toit_rest_api.toit
core_utils.toit
ble_utils.toit
ble_helper.toit
TOIT
colmi_rx
ble_utils.toit
clear.toit
colmi_runner_db.toit
colmi_runner.toit
colmi_rx_04012026.txt
core_utils.toit
package.lock
package.yaml
periodic_timer.toit
storage_destroy.toit
storage_init.toit
storage_utils.toit
toit_rest_api.toit
crypt
crsa.toit
crypto_utils.toit
custom_rsa.toit
histoty.txt
hybrid_client.toit
hybrid_server.toit
rsa_decrypter.toit
rsa_encrypter.toit
rsa_helper.toit
runner.toit
OUTLINE
TIMELINE
impManualMeasure value/ByteArray -> any :
subtype := value[1]
error_code := value[2]
response := value[3] & 0xff
systolic := value[4] & 0xff
if subtype == TEMPERATURE :
response = response < 11 ? 0 : response.to_float/10 + 20
else :
if subtype == FATIGUE :
response = response >> 1
else :
if subtype == BLOOD_PRESSURE :
response = "$systolic/$response"
else :
if subtype == BLOOD_SUGAR :
response = (18*response.to_float/10).round
if error_code == 0 :
return "$response $(parameter-unit subtype)"
else if error_code == 1 :
print "Live $(parameter-name subtype) error code $error_code received from ring"
return "Measure error"
else if error_code == 2 :
return "Missing data"
else :
return "Unknown Measure error"
PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL
micrcx@micrcx-desktop:~/toit$ cd ./colmi_rx
micrcx@micrcx-desktop:~/toit/colmi_rx$ export PATH=$HOME/jaguar/:$PATH
micrcx@micrcx-desktop:~/toit/colmi_rx$ jag -d basic run colmi_runner_db.toit
Scanning for device with name: 'basic'
Running 'colmi_runner_db.toit' on 'basic' ...
Success: Sent 144KB code to 'basic' in 4.56s
micrcx@micrcx-desktop:~/toit/colmi_rx$
Ln 123, Col 41 Spaces: 2 UTF-8 LF { } Toit Finish Setup


Jan 4 10:13
Google Keep
auth - Realtime Database
console.firebase.google.com/project/auth-2b7d3/database/auth-2b7d3-default-rtdb/data
Google Chrome isn't your default browser
Set as default
auth
Realtime Database
Need help with Realtime Database? Ask Gemini
Data Rules Backups Usage Extensions
Protect your Realtime Database resources from abuse, such as billing fraud or phishing
Configure App Check
https://auth-2b7d3-default-rtdb.firebaseio.com
rssi: -82
spo2: "97 %"
time: "2026/01/04 08:02:44.177"
-Oi71CMsk2pPEgGSzSUH
battery: "95% [false]"
ca: 6
mac: "30:38:47:31:08:03"
name: "R09_0803"
rssi: -88
temp: "35.700000000000002842 °C"
time: "2026/01/04 08:03:57.687"
-Oi71QfZzcM-3ApUkUfB
battery: "95% [false]"
Database location: United States (us-central1)

Measurement duration

Good question. Measurement duration depends on the type of measurement. In my experience, it ranges from 30 seconds for heart rate to 1 minute for HRV, stress, or blood sugar. So, measuring all seven parameters can take 3 to 5 minutes.

It's not that simple

The image of the measurement results above is worth paying attention to. We have two rings measuring the same person at the same time, yet they yield different results. I included the images intentionally, not so much to highlight the imperfections of the devices, but to show that the measurement results are a matter of trust and almost religious faith. If there were only one ring, there wouldn't be any questions. Incidentally, I'm sure discrepancies would inevitably appear on the **Oura** and **Samsung** reference rings as well.

Can I trust the measurement results?

Most measurements in wearable devices use an optical *PPG* (photoelectric plethysmography) sensor with green, red, and infrared LEDs. It measures *heart rate*, *blood oxygen saturation* (*SpO_2*), and *blood pressure*, and also helps monitor sleep (sleep stages and quality) and sometimes *stress* (based on heart rate variability (*HRV*). A skin *temperature* sensor is also used to monitor *body/skin temperature*.

Heart rate, *blood oxygen*, *temperature*, and even *blood pressure* data can be trusted if measurements are taken in a state of relative rest. There are deviations in data obtained by medical pulse oximeters, but they are quite acceptable. With *temperature* changes, the discrepancy can reach ~0.4–1°C, compared to a regular thermometer.

What definitely shouldn't be credit are the *HRV* and *stress* data. They vary greatly and can easily be classified as "*bullshit*".

Blood Sugar/Glucose

COLMI models don't explicitly advertise the ability to measure *blood glucose* levels, although the option is indeed available. Incidentally, many *Chinese brands* claim that *"non-invasive" glucose measurement* using a *PPG* sensor (the same type as the *Vcare VC30F in COLMI*) is "possible":

- Light (green/red/IR) passes through the skin and reflects off the blood.
- The algorithm attempts to indirectly estimate glucose based on changes in light absorption, pulse wave, or HRV (heart rate variability).
- Sometimes they add the now-fashionable term "AI" or simple formulas for "trend estimation".

So, if results HRV and stress measurements are "*bullshit*", then blood sugar/glucose data obtained with a PPG sensor is "*complete and utter bullshit*".

The US FDA and similar regulators (as of 2026) have not approved any smart rings or watches for non-invasive glucose monitoring. All such devices are either marketing hype or experimental with low accuracy.

Myths about the Apple Watch's blood sugar/glucose measurement capabilities

Currently (January 2026), the ***Apple Watch*** doesn't have built-in blood glucose measurement. No model, including *Series 10*, *Series 11*, *Ultra 2*, or *Ultra 3*, is equipped with a sensor for non-invasive (puncture-free) *blood sugar* monitoring.
Apple has been working on this technology for over **15 years** (since the Steve Jobs era), using optical methods (lasers and spectroscopy to analyze interstitial fluid). However, according to recent reports (*Bloomberg*, *MacRumors & etc*):

- The feature is “still many years away” from release (March 2025 report).
- Possible debut: no earlier than 2027 (Apple Watch Series 13) or later (2028+).
- Even then, these may only be trends and prediabetes warnings, not accurate medical measurements (due to difficulties with accuracy and miniaturization).

All of the above is why I don't want to hand links to the created projects in the git repositories in this case. I'd really hate to hand over to enthusiasts a tool that could potentially detect symptoms of every disease in the world except childbed fever (*“Three Men in a Boat” by Jerome K. Jerome*).

The only links I'm providing are: *Issue 5573* – a link to an issue in *Gadgetbridge* that explains how to measure the stated parameters of the ring. And a link to my old post, which mentions measuring blood sugar levels with yet another *Chinese Wunderwaffe*. In the case of the ring, we're dealing with a similar product.

Summary

So, to summarize all of the above, we can draw the following conclusion:

1. The *smart ring form factor is almost ideal* as a lightweight sensor for vital medical information. Only a sensor implanted in the human body can compete with it. I hope, however, that the inquisitive human mind won't have time to implement this revolutionary idea.

2. The presence of a *BLE interface allows* for remote measurement of required parameters, s*toring them in the cloud*, and subsequent analysis. This is, of course, provided that this data reflects reality and can be trusted, and that someone needs it.

3. In other words, the creation of an infrastructure for mass monitoring of population health is *already technically possible*. It's just a matter of progress in the field of measurement.